

SYDNEY OBSERVATORY, Australia

WMO: 947680

Lat: 33.8608S

Lon: 151.2050E

Elev: 40

StdP: 100.84

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	8.0	-2.8	3.0	16.1	-1.4	3.4	15.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.0	31.7	19.8	29.3	20.1	27.7	20.2	23.0	27.4	22.4	26.4	21.8	25.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.4	24.9	21.1	15.8	24.4	20.5	15.3	24.1	68.5	27.6	66.1	26.4	64.0	25.8	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	5.6	37.7	0.7	3.1	5.1	40.0	4.7	41.8	4.3	43.6	3.8	45.8
			WB	3.2	24.0	0.7	0.9	2.7	24.7	2.2	25.2	1.8	25.7	1.3	26.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	23.1	23.0	21.8	19.2	16.4	14.0	13.3	14.3	16.6	18.5	20.0	21.7
	DBStd	4.12	2.34	2.13	2.05	2.15	2.07	1.82	1.81	2.20	2.69	2.91	2.76	2.49
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	611	0	0	1	14	67	131	157	128	66	32	12	2
	CDD10.0	3092	406	364	367	275	198	119	102	132	199	264	301	364
	CDD18.3	661	147	131	110	39	7	0	1	2	14	38	63	108
	CDH23.3	2922	734	623	372	107	13	0	1	11	92	216	286	468
	CDH26.7	648	168	111	48	16	0	0	0	1	22	73	89	120
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1253	119	125	141	120	113	125	65	95	65	87	106	84
	PrecMax	2036	385	631	388	507	371	408	182	483	226	285	517	279
	PrecMin	625	6	10	8	5	8	4	2	0	3	1	10	3
	PrecStd	334	82	118	97	109	87	98	50	109	57	79	105	66
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	33.0	30.6	28.9	25.2	21.3	22.2	24.7	29.4	32.8	33.8	33.9
		MCWB	20.3	22.7	20.5	17.9	16.0	14.6	13.5	13.7	16.0	17.0	18.8	19.0
	2%	DB	30.4	29.3	27.6	25.7	22.8	19.5	19.5	21.9	26.0	28.4	29.0	29.7
		MCWB	21.0	21.9	20.7	18.1	15.9	13.9	12.2	13.3	14.9	17.0	18.7	19.9
	5%	DB	28.0	27.7	26.4	24.3	21.4	18.4	18.0	20.1	23.3	25.3	26.2	27.3
		MCWB	21.5	21.6	20.3	18.0	15.5	13.4	11.9	12.9	14.4	16.6	18.9	20.1
	10%	DB	26.7	26.5	25.3	23.1	20.3	17.4	16.8	18.6	21.4	23.2	24.3	25.6
		MCWB	20.9	21.0	19.8	17.5	15.2	13.1	11.4	12.3	14.4	16.3	18.3	19.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	24.1	22.7	20.7	18.9	16.5	15.7	16.4	18.2	20.1	21.9	22.8
		MCDB	28.4	28.6	26.6	24.4	21.0	19.2	19.1	20.7	23.9	25.7	27.3	27.5
	2%	WB	22.8	23.2	22.0	19.9	17.9	15.5	14.2	15.2	17.2	19.0	20.7	21.8
		MCDB	27.4	27.3	25.3	23.6	21.0	17.8	16.9	19.1	21.9	24.2	25.4	26.5
	5%	WB	22.2	22.6	21.4	19.2	17.0	14.7	13.3	14.3	16.4	18.2	20.0	21.2
		MCDB	26.3	26.4	24.8	22.7	19.9	17.1	16.1	18.2	20.9	22.7	24.2	25.3
	10%	WB	21.7	21.9	20.8	18.5	16.1	14.0	12.5	13.4	15.6	17.5	19.3	20.5
		MCDB	25.5	25.5	24.2	21.9	19.2	16.6	15.5	17.3	19.9	21.7	23.1	24.4
Mean Daily Temperature Range	5% DB	MDBR	6.0	5.9	6.2	7.3	7.6	7.0	7.9	8.4	8.4	7.7	6.7	6.4
		MCDBR	8.9	7.8	7.9	9.0	9.1	8.5	9.5	10.4	11.7	11.9	10.8	9.9
		MCWBR	3.5	3.1	3.4	3.9	4.4	4.5	4.7	4.9	4.8	4.8	4.2	3.8
	5% WB	MCDBR	7.2	7.0	6.5	7.2	7.1	6.7	7.6	8.4	8.9	8.6	8.3	7.7
		MCWBR	3.2	3.2	3.3	3.5	4.0	4.1	4.5	4.9	4.7	4.1	3.7	3.5
Clear-Sky Solar Irradiance	taub	0.394	0.395	0.359	0.333	0.319	0.311	0.301	0.310	0.343	0.361	0.384	0.395	
	taud	2.463	2.478	2.567	2.616	2.603	2.621	2.631	2.599	2.501	2.482	2.456	2.437	
	Ebn at Noon	944	922	921	892	852	834	865	905	921	944	947	946	
	Edh at Noon	118	113	96	82	73	68	70	81	101	111	118	122	
All-Sky Solar Radiation	RadAvg	6.50	5.62	4.77	3.83	3.00	2.37	2.78	3.68	4.90	5.70	6.11	6.64	
	RadStd	0.46	0.48	0.41	0.32	0.21	0.22	0.22	0.20	0.27	0.47	0.54	0.52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		+0.46	+0.21	N/A	+0.74	N/A	N/A	N/A	-78	+167	+84
Regional (33 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-1	-86	+187	+98

Nomenclature: See separate page